

Ankit Sharma

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EDUCATION

VIT Bhopal University

B.Tech in Computer Science and Engineering (CGPA: 8.80 /10)

Bhopal, India

Jun 2023 – Jun 2027

Jawahar Navodaya Vidyalaya

Senior Secondary Education (CBSE, Class XII: 84.4%, Class X: 92.8%)

Bilaspur, India

2020 – 2022

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, SQL, HTML/CSS, JavaScript

AI & Machine Learning: Generative AI, LLMs, RAG, Computer Vision (YOLOv8, CNNs), NLP, Transfer Learning, Deep Learning

Frameworks & Tools: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, LangChain, OpenCV, NumPy, Pandas, FastAPI, Flask, Streamlit, AWS, Git/GitHub, Render, Vercel, Docker

CS Fundamentals: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, Data Engineering

EXPERIENCE

Coding Jr | AI Research Intern | [Certificate](#)

Apr 2025 – Jul 2025

- Spearheaded research into transformer-based LLMs to architect an AI coding agent, mitigating context degradation by evaluating Code-RAG pipelines with Abstract Syntax Tree (AST) chunking and a 10%–20% sliding window overlap.
- Evaluated semantic prompt compression frameworks—including LongLLMLingua (achieving 4x compression) and CodePromptZip—to maximize token efficiency, reduce inference latency, and systematically eliminate syntax hallucination.
- Synthesized architectural blueprints for on-the-fly repository navigation, comparing 4-bit quantized QLoRA against ReAct agentic workflows to enforce strict syntactic adherence for production code generation.

PROJECTS

1. Heal Bridge: AI Medical Triage Platform | [GitHub](#) | [Live](#)

Hybrid AI Architecture & Serverless Inference

Python, LLMs, FastAPI, Scikit-Learn, Three.js

- Architected a serverless medical triage platform to eliminate diagnostic hallucinations by decoupling Meta Llama-3 (Groq LPU)'s generative NLP from a Bernoulli Naive Bayes classifier trained on 1M+ patient records.
- Engineered a multimodal pipeline utilizing a 4-model Google Gemini chain for clinical OCR and live camera analysis, mapping inputs into a 959-column sparse matrix via PubMedBERT vector search.
- Deployed a stateless Vercel and Hugging Face architecture, utilizing a Llama-3 agent to synthesize bilingual reports while integrating a Three.js 3D anatomical mapper and an Overpass API geospatial router.

2. Real-Time Edge Biometric Gait Recognition | [GitHub](#)

Real-Time Computer Vision & Edge Inference

PyTorch, OpenCV, Transfer Learning, YOLOv8

- Architected an edge biometric pipeline identifying walking patterns, processing 2M+ pixels per frame via 1080p sensors (60% ROI) to eliminate cloud latency.
- Engineered a multithreaded OpenCV and YOLOv8 pipeline to isolate binary human silhouettes—strictly utilizing silhouette geometries rather than skeletal/joint mapping—feeding a PyTorch Xception for zero-latency inference.
- Resolved training-serving skew by patching a live 64x64 tensor distortion bug, utilizing temporal majority voting to achieve 98% accuracy on CASIA-B (124 subjects, 11 angles).

3. Brain Tumor Detection System | [GitHub](#)

Deep Learning Image Classification

TensorFlow, Transfer Learning, Flask, CNN

- Engineered a dual-input diagnostic pipeline processing tabular clinical data alongside 3,000+ MRI scans (Brats2022 dataset) to automate tumor detection.
- Trained custom CNNs and transfer learning models (VGG16/ResNet50) using CLAHE enhancement and dropout to prevent overfitting.
- Deployed a unified Flask REST API for real-time inference, achieving 96%+ test accuracy, a 0.94 weighted F1-score, and 0.95 precision.

LEADERSHIP & EXTRACURRICULARS

Non-Technical Clubs VIT Bhopal | Core Team Member

Mar 2025 – Oct 2025

KHO-KHO | Regional Level

ACHIEVEMENTS & CERTIFICATIONS

- **Competitive Programming** – Solved 280+ Data Structures and Algorithms problems on LeetCode using C++ [Profile](#)
- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** [Certificate](#)
- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** [Certificate](#)
- **The Bits and Bytes of Computer Networking** – Google (Coursera) [Certificate](#)
- **Introduction to Machine Learning (Elite)** – NPTEL (Swayam) [Certificate](#)
- **Marketing Analytics (Elite+Gold)** – NPTEL (Swayam) [Certificate](#)